

LING 411 Problem Session: Designing a Judgment Experiment

Reading: Schütze & Sprouse, Judgment data (Podesva & Sharma, Ch. 3)

Outline (45 min). Part 1: evaluating three designs (10 min). Part 2: designing an experiment in groups of 3–4 (25 min). Part 3: peer review (10 min).

Steps in designing a judgment experiment

Step	Notes
1. Phenomenon	State your hypothesis H and an alternative H'. They should predict different ratings in at least one condition.
2. Conditions	The factors are the independent variables . Two factors with two levels each give 4 conditions. Check the main effects and the interaction.
3. Task	The responses are the dependent variable . Forced choice shows which sentence is better; Likert scale and magnitude estimation show how much better; yes-no gives proportions.
4. Items	Write each item in all conditions. The versions should differ only in the factors.
5. Confounds	Lexical, semantic, structural, pragmatic, speaker-related.
6. Counterbalancing	Latin square: with 4 conditions, make 4 lists. Each participant sees each item once, in one condition.
7. Fillers and practice	Filler ratio 1:2 or 1:3. Fillers should vary in structure and acceptability. Practice items show the low, middle, and high points of the scale.

Part 1. Evaluating designs

Each design below has a problem. Identify the problem and propose a solution.

Design 1. A study tests whether English speakers accept preposition stranding (*Who did you talk to?*). Naive participants are asked: “Is this sentence grammatical?”

Design 2. A study tests the *that*-trace effect by comparing *Who do you think left?* with *Who do you think that the extremely tired runners congratulated?*

Design 3. A 2×2 study has 16 items. Every participant rates all 4 versions of all 16 items (64 sentences). There are no fillers.

Part 2. Designing an experiment

Choose one of the phenomena below, or another phenomenon you are interested in, and complete the design sheet.

- A. Agreement attraction.** *The key to the cabinets are rusty* is ungrammatical, but speakers often rate it higher than *The key to the cabinet are rusty*. Does a plural noun near the verb affect judgments?
- B. Superiority.** *Who bought what?* sounds better than *What did who buy?* With *which*-phrases (*Which book did which student buy?*) the difference is reported to be smaller.
- C. Comparative illusion.** *More people have been to Russia than I have.* Many speakers first judge this sentence acceptable, but it has no clear meaning.
- D. NPI illusion.** *No bills that the senators voted for will ever become law* is fine, and *The bills that the senators voted for will ever become law* is not. What about *The bills that no senators voted for will ever become law*?

Design sheet

1. H and H'	
2. Factors and levels. Where do H and H' differ?	
3. Task and instruction. Why this task?	
4. One item in all conditions	
5. Two possible confounds and how to control them	
6. Number of items and lists	
7. Fillers and practice items	

Part 3. Peer review

Exchange design sheets with another group. Write one strength of their design and one point that needs to be revised. Consider these questions: Do H and H' predict different results? Do the conditions differ only in the factors? Does the task provide the information needed to test H? Could participants guess the purpose of the study?